

TRINETRA

AI-Powered Email Threat Detection, GeoLocation & Forensic Intelligence Platform

Smart India Hackathon 2026

Problem Statement: SIH26106

Team: TrinetraAI

1. Executive Summary

TRINETRA is a prototype email-security and forensic-intelligence platform designed around SIH26106. It combines email artifact parsing, authentication/header analysis, behavioral NLP, URL and domain analysis, IP/network intelligence, geolocation, graph correlation, risk fusion, case history and structured forensic reporting. The prototype demonstrates artifact-based analysis; true pre-delivery blocking requires integration at a mail gateway, SMTP/IMAP/API or equivalent institutional control point.

2. SIH26106 Problem Context

The problem statement calls for detection of fraudulent email together with technical origin tracing and forensic intelligence. Its requirements include NLP/ML detection, header/protocol analysis, SPF/DKIM/DMARC, relay reconstruction, IP/geolocation analysis, anonymizer indicators, domain intelligence, threat-intelligence correlation, graph analysis, alerts, forensic reports, case management, and privacy/evidence safeguards.

3. Research Basis and Gap

A 2023 systematic literature review examined 38 BEC-phishing studies from 950 initially identified papers and notes the difficulty of detecting many BEC messages with conventional approaches. It also reports frequent use of body and header features. Research on malicious-site geolocation describes IP-based passive geolocation as a rapid, database-driven approach whose reliability varies by database. TRINETRA combines technical email evidence, content/behavioral signals, infrastructure context, historical correlation and graph intelligence in one investigation workflow.

4. Proposed Solution

Raw Email → Preserve & Seal → Parse → Analyze → Enrich → Fuse Evidence → Correlate → Investigate → Report. Primary users include SOC analysts, incident responders, institutional security teams, cybercrime/fraud investigation workflows and enterprise/government security administrators.

5. End-to-End Architecture

The architecture contains ingestion/evidence preservation, RFC 5322/MIME processing, header/authentication analysis, content/behavioral analysis, URL/domain intelligence, GeoIP/ASN context, multi-layer risk fusion, graph intelligence, attribution support and structured forensic reporting.

6. Privacy, Evidence Integrity and Chain of Custody

The original artifact is preserved before downstream analysis. SHA-256 sealing and later verification provide integrity checks; sensitive information can be masked for display and processing. Public repositories must not contain private emails, credentials, API keys or unnecessary personal data.

7. Prototype Interface

The prototype demonstrates an active investigation workspace, three-layer evidence package, NLP behavioral analysis, header authentication, approximate-origin intelligence, graph intelligence, URL/attachment findings, searchable case history and a structured forensic PDF.

8. Demonstrated Scenarios

Scenario 1: Executive BEC Fraud. Scenario 2: Banking/KYC phishing. Scenario 3: Clean Corporate Notice. These are controlled demonstration scenarios and are not evidence about real organizations or domains.

9. Testing and Verification

The repository documentation reports 11 backend tests passing, including RFC 5322 parsing, SPF/DKIM-related analysis, risk fusion and PDF export, plus a successful frontend production build. A production evaluation should additionally report precision, recall, F1, false-positive/negative rates, latency, throughput and geolocation error/confidence using a reproducible dataset.

10. Technical Stack

Frontend: React 19, Vite, Tailwind CSS, Leaflet. Backend: Python 3.11+, FastAPI. Security: RFC 5322/MIME, SPF/DKIM/DMARC, URL/domain analysis. Intelligence: MaxMind GeoLite2 City/ASN and optional external feeds. AI/NLP: Gemini where configured plus deterministic offline fallback. Data/Graph: SQLite, optional PostgreSQL/Neo4j. Reporting: structured forensic PDF generation.

11. Real-Time / Pre-Interaction Deployment

The SIH requirement calls for real/near-real-time alerting before user interaction. The current prototype demonstrates artifact ingestion and analysis. A production deployment would place TRINETRA at a mail gateway, SMTP/IMAP/provider API or equivalent control point, producing a decision such as deliver, quarantine or alert before normal user access.

12. Feasibility

Technical feasibility follows from established web, parsing, intelligence and database components. Operationally, the dashboard is designed for analyst workflows. Local GeoIP and deterministic analysis reduce dependence on remote services. The pipeline can be separated into workers/services for higher-volume deployment.

13. Key Engineering Advantages

Evidence-first processing; multi-layer evidence fusion; offline-capable local analysis; campaign-level graph correlation; and explainable investigation views.

14. Limitations

GeoIP is approximate; VPN/Tor/proxy/compromised accounts can obscure origin; header inference depends on available data; AI outputs are probabilistic; external feeds can be rate-limited; artifact ingestion is not the same as production interception; and a generated legal certificate does not itself guarantee court admissibility.

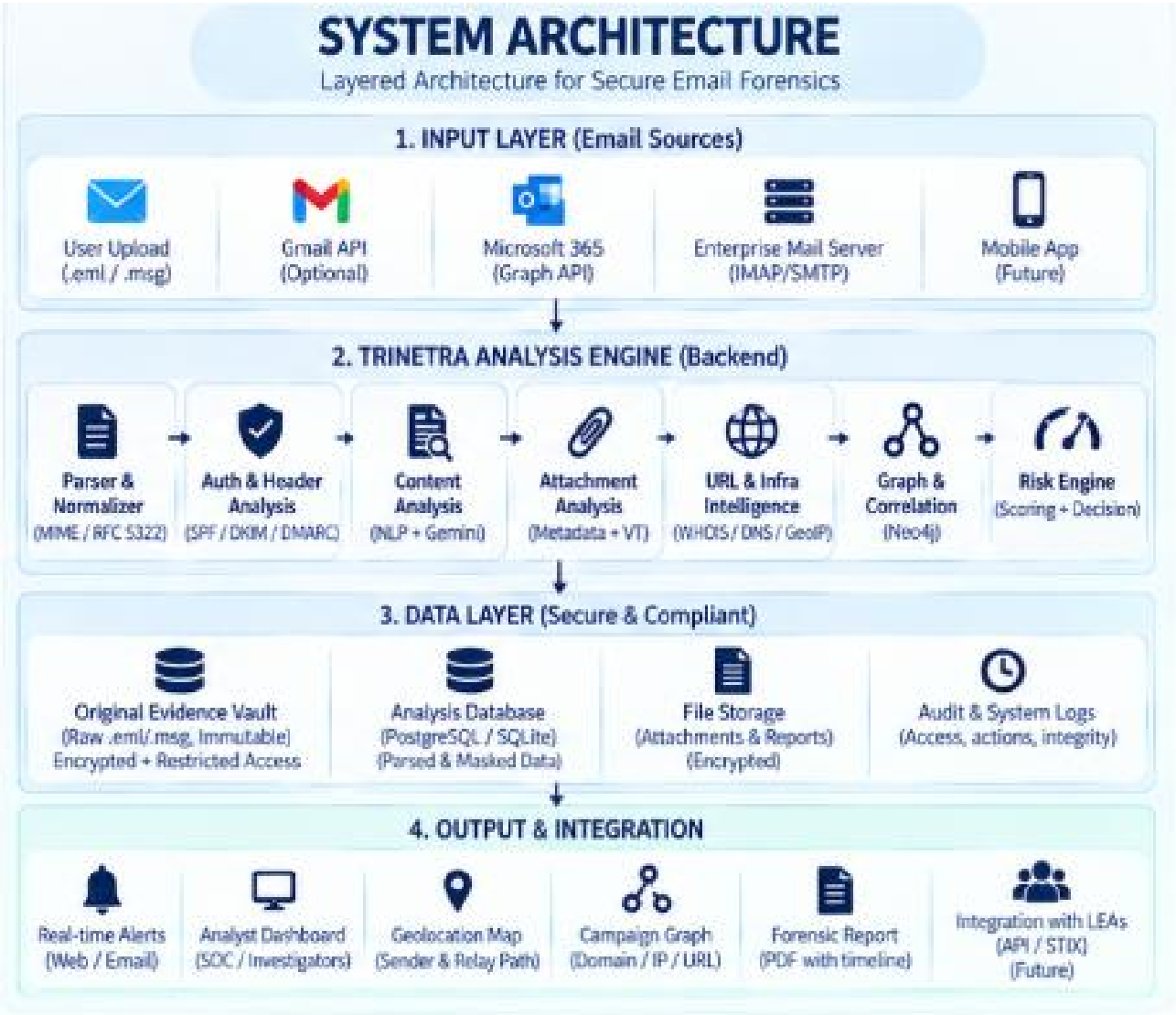
15. Future Scope

Mail-gateway integration, enterprise provider connectors, SIEM/SOC integration, richer DNS/WHOIS intelligence, additional threat feeds, calibrated ML models, campaign clustering, advanced attachment/media analysis, analyst feedback and scalable distributed processing.

16. Conclusion

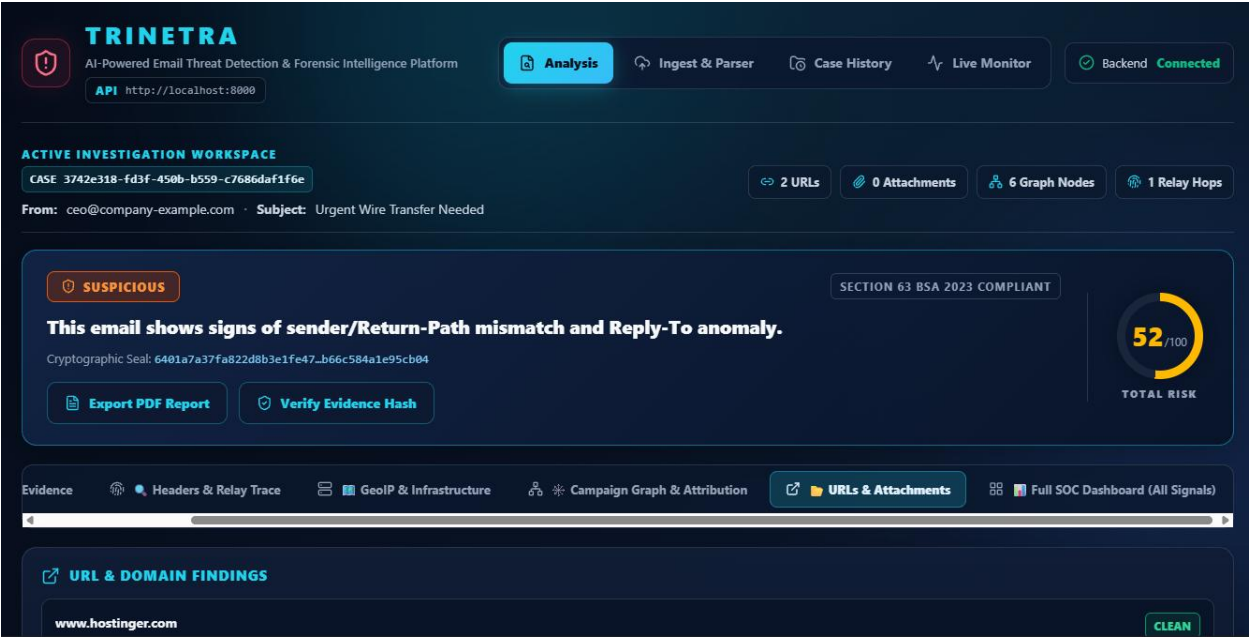
TRINETRA integrates email evidence, behavioral signals, infrastructure context and relationship-level intelligence into one analyst workflow. The prototype demonstrates the journey from email artifact to risk assessment, approximate origin context, graph correlation, case history and structured reporting.

Architecture Reference



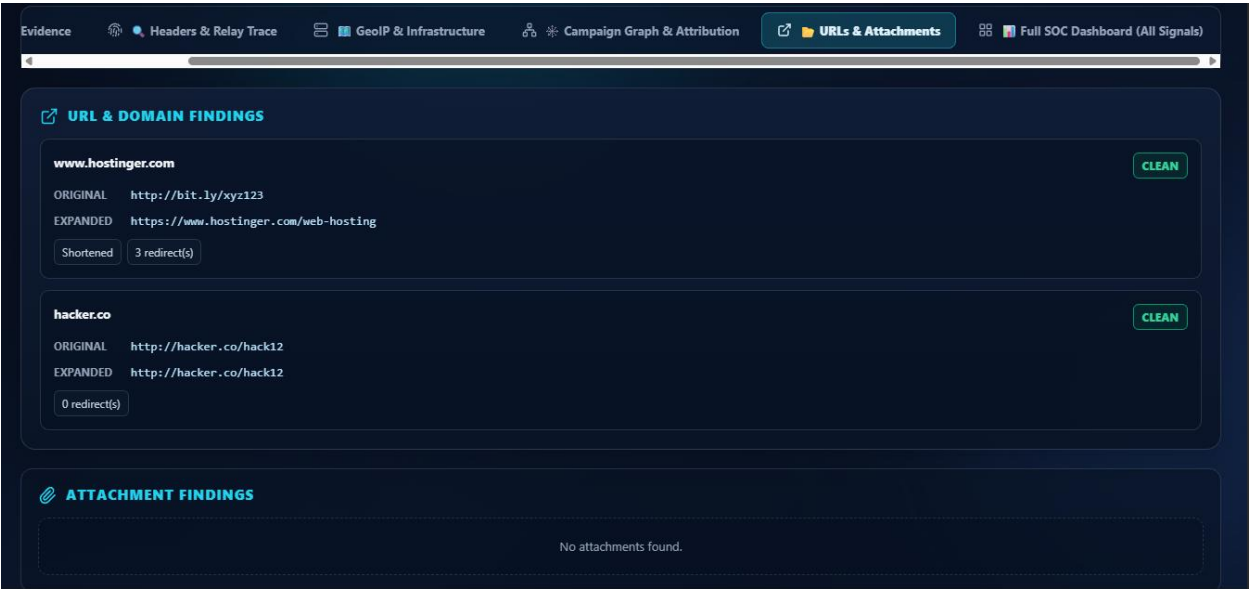
17. Prototype Visual Evidence

Active Investigation Workspace



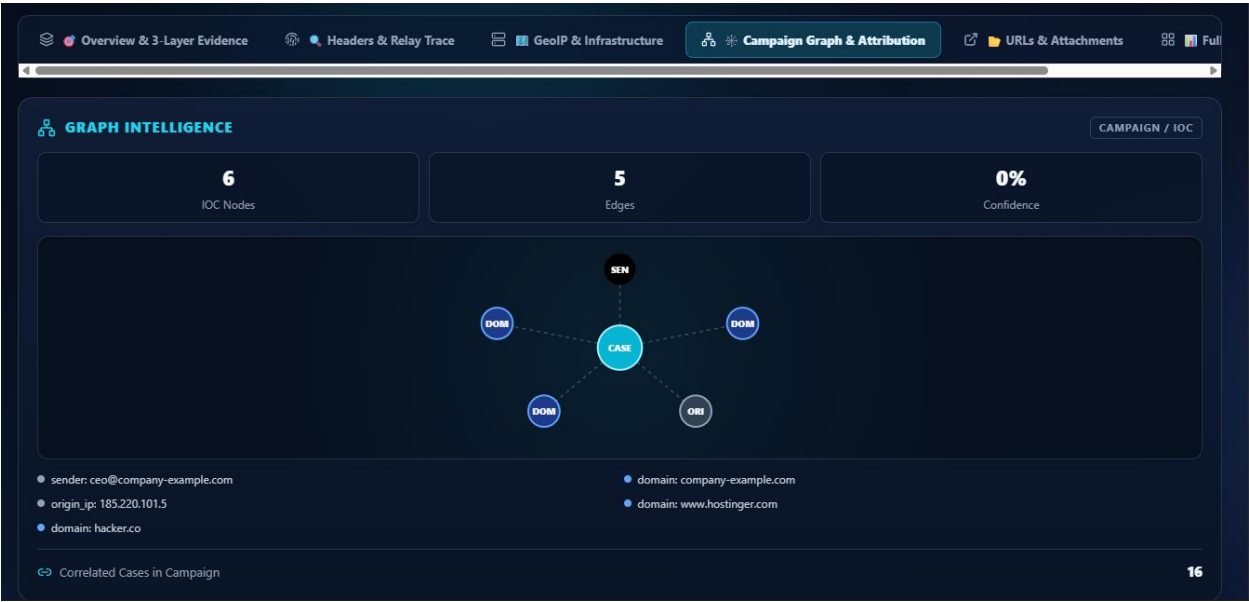
Prototype screenshot supplied for project documentation.

Three-Layer Evidence Fusion



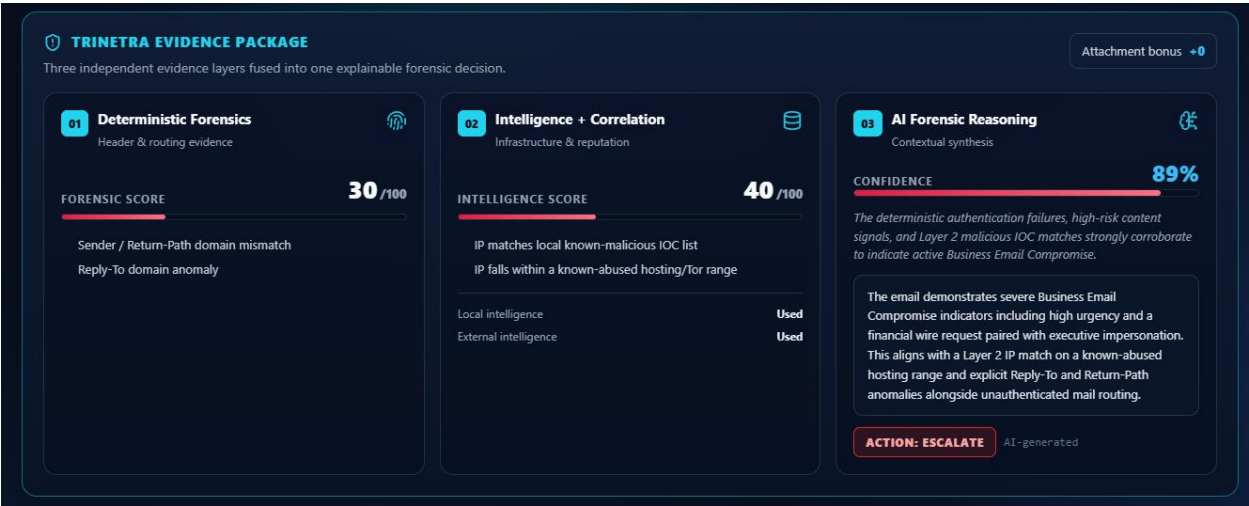
Prototype screenshot supplied for project documentation.

NLP Behavioral Analysis and Attribution



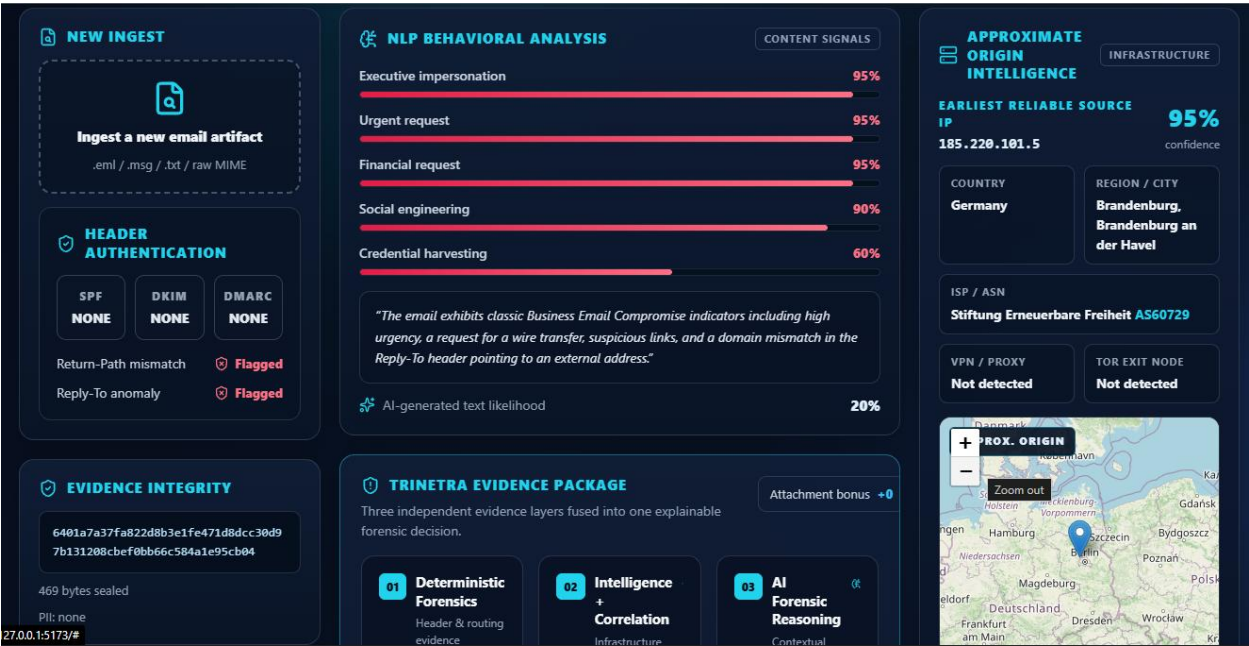
Prototype screenshot supplied for project documentation.

Graph Intelligence



Prototype screenshot supplied for project documentation.

URL and Attachment Findings



Prototype screenshot supplied for project documentation.

Forensic Case History

Forensic Case History					
Archived email investigations stored in local runtime database. Select any case to resume the forensic workspace.					
Timestamp ↓	SENDER IDENTITY	SUBJECT LINE	Risk	CLASSIFICATION	ACTIONS
26, 12:07:04	ceo@company-example.com 3742e318-fd3f-45...	Urgent Wire Transfer Needed	52/100	SUSPICIOUS	Open Case
26, 7:01:10	ceo@company-example.com b8d59935-asd7-41...	Urgent Wire Transfer Needed	54/100	SUSPICIOUS	Open Case
26, 6:40:54	— 287c0eba-87b8-42...	—	21/100	LOW RISK	Open Case
26, 5:10:33	ceo@company-example.com 317ce1f6-a5f9-44...	Urgent Wire Transfer Needed	54/100	SUSPICIOUS	Open Case
26, 12:37:21	alice@company.com 1b439d94-8e18-47...	Project Update Meeting	2/100	SAFE	Open Case
26, 12:36:45	ceo@company-example.com cc3b5b77-87e7-42...	Urgent Wire Transfer Needed	53/100	SUSPICIOUS	Open Case

Prototype screenshot supplied for project documentation.

Forensic Technical Report

TRINETRA AI

Email Threat Detection and Forensic Intelligence

Case / Reference ID	3742e318-fd3f-450b-b559-c7686daf1f6e
Report Type	Forensic Technical Examination Report
Generated	2026-09-20 09:32:23 UTC
Prepared By	Trinetra AI forensic analysis system

Purpose: This report records technical findings generated from the supplied email artifact and associated analysis evidence. It is intended to assist investigators, law-enforcement personnel and legal professionals.

Important: This is a technical forensic report, not a judicial finding of guilt, identity or attribution. Final legal characterization and evidentiary weight are determined by the competent investigating authority and court.

Legal and Evidentiary Framework

Framework	Application to this report
Bharatiya Sakshya Adhinyam, 2023 - Sections 61 to 63	Electronic or digital records are recognized as evidence subject to the statutory requirements. Section 63 addresses admissibility of electronic records and the certificate requirements for computer output.
Information Technology Act, 2000	Potentially relevant where the investigated conduct satisfies the elements of an applicable provision, including Sections 43, 66, 66C or 66D. The exact offence must be determined from the facts and by the investigating authority; this report does not automatically charge an offence.
Other applicable criminal law	The investigating authority may identify applicable provisions of the Bharatiya Nyaya Sanhita, 2023 or other special legislation based on the facts of the case.

Prototype screenshot supplied for project documentation.

Live Monitor

 AI-Powered Email Threat Detection & Forensic Intelligence Platform

 Analysis  Ingest & Parser  Case History  **Live Monitor**  Backend **Connected**

API <http://localhost:8000>

 **Live Mailbox Monitor** STOPPED

Optional automated ingestion path: unseen IMAP messages are sent through the same forensic pipeline and can create high-risk cases before manual interaction.

Mailbox username

App password

IMAP server

Folder

imap.gmail.com

INBOX

Poll seconds

20

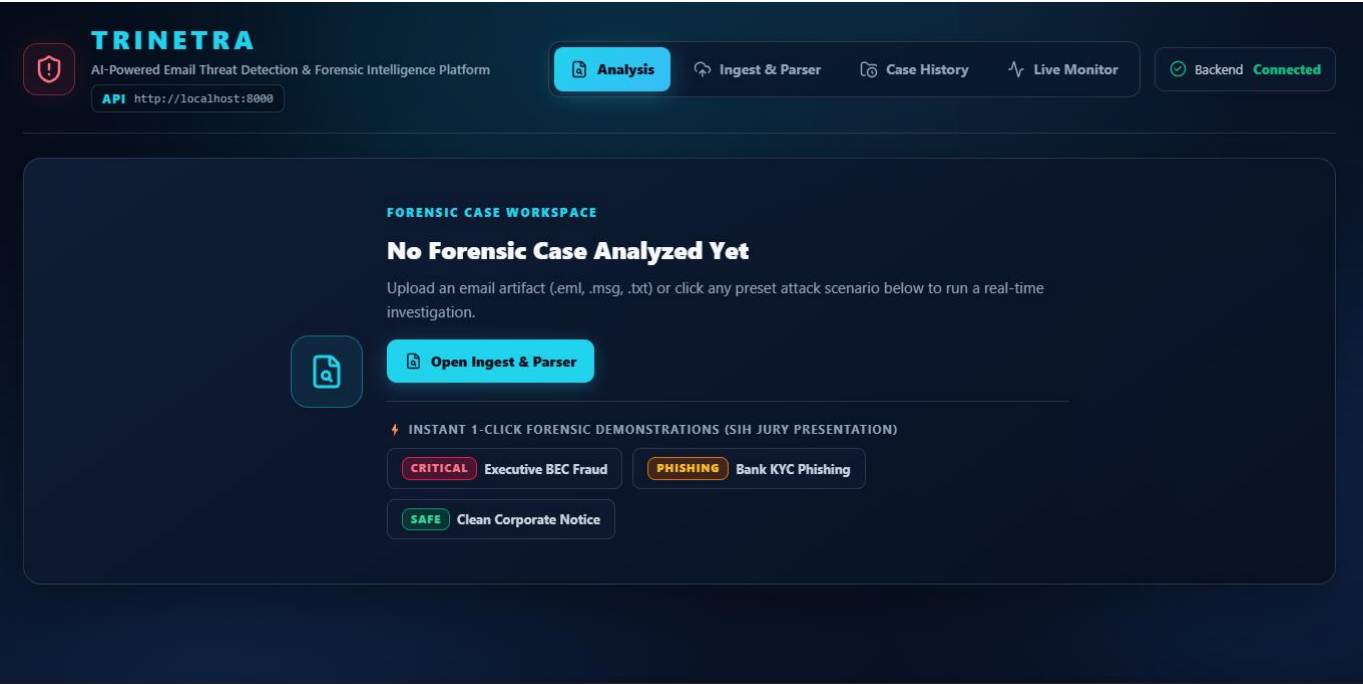
 Start monitor  Stop 

Automatically processed: 0

HIGH-RISK ALERTS • 0

No high-risk alerts returned.

Prototype screenshot supplied for project documentation.



18. References

1. Atlam, H. F., & Oluwatimilehin, O. (2023). Business Email Compromise Phishing Detection Based on Machine Learning: A Systematic Literature Review. *Electronics*, 12(1), 42. <https://doi.org/10.3390/electronics12010042>
2. Mansoori, M., & Welch, I. (2020). How do they find us? A study of geolocation tracking techniques of malicious web sites. *Computers & Security*, 97, 101948. <https://doi.org/10.1016/j.cose.2020.101948>
3. Geolocation Based E-Mail Forensics: A Timeline Analysis Approach. *IEEE Xplore*, document 11263637.
4. Phishing in India – Analytical Study. *ResearchGate*.
5. Government of India. The Bharatiya Sakshya Adhinyam, 2023, Section 63.
6. Smart India Hackathon 2026 — Problem Statement SIH26106.

Report note: This document distinguishes demonstrated prototype functionality from future production integration and does not treat automated analysis or generated reports as proof of human identity or guaranteed legal admissibility.